01. Let $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}_{2\times 3}$, $B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 3 & 1 \end{bmatrix}_{3\times 2}$ and $C = \begin{bmatrix} 1 & 1 \\ 2 & -2 \end{bmatrix}_{2\times 2}$

Find $A \cdot B + C^T$:

* $A \cdot B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}_{2\times 3} \times \begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 3 & 1 \end{bmatrix}_{3\times 2} = \begin{bmatrix} 7 & 8 \\ 13 & 14 \end{bmatrix}_{2\times 2}$

$$A \cdot B + C^T = \begin{bmatrix} 7 & 8 \\ 13 & 14 \end{bmatrix}_{2\times 2} + \begin{bmatrix} 1 & 2 \\ 1 & -2 \end{bmatrix}_{2\times 2}$$

$$= \begin{bmatrix} 8 & 10 \\ 14 & 12 \end{bmatrix}_{2\times 2}$$

Ans

a. $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}_{2\times 3}$ and $B = \begin{bmatrix} 2 & 3 & 2 \\ 1 & 2 & 2 \end{bmatrix}_{2\times 3}$

* $A \cdot B^T = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}_{2\times 3} \times \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 2 & 2 \end{bmatrix}_{3\times 2}$

$$= \begin{bmatrix} 10 & 7 \\ 11 & 8 \end{bmatrix}_{2\times 2}$$

properties of Matrix multiplication

* $A \cdot B \neq B \cdot A$
* $A \cdot (B \cdot C) = (A \cdot B) C$
* $I \cdot A = A \cdot I = A$
* $A \cdot (B + C) = A \cdot B + B A C$

Determinant of a Matrix

if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2\times 2}$ then $|A| = a\times d - b\times c$ = ?

ex① if $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}_{2\times 2}$, $|A| = 2\times 4 - 1\times 3 = \underline{5}$

② if $A = \begin{bmatrix} 2 & -3 \\ 4 & 5 \end{bmatrix}_{2\times 2}$, $|A| = 2\times 5 - 4\times(-3) = \underline{22}$

Inverse of a Matrix

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2\times 2}$ is

$$A^{-1} = \frac{1}{|A|}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}_{2\times 2}$$

Q. 01. $A = \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix}_{2\times 2}$ Find A^{-} :-

$|A| = 2\times 1 - 3\times 1 = (-1)$

$$A^{-} = \frac{1}{|A|}\begin{bmatrix} 1 & -1 \\ -3 & 2 \end{bmatrix} = \frac{1}{-1}\begin{bmatrix} 1 & -1 \\ -3 & 2 \end{bmatrix} = (-1)\begin{bmatrix} 1 & -1 \\ -3 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & +1 \\ 3 & -2 \end{bmatrix}_{2\times 2}$$

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01. $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \end{pmatrix}_{2\times 3}$ $B = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}_{2\times 2}$ $C = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}_{2\times 3}$

a) $A \cdot B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \end{pmatrix}_{2\times 3} \times \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}_{2\times 2}$ = not defined

b) $B \cdot A = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}_{2\times 2} \times \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \end{pmatrix}_{2\times 3} = \begin{pmatrix} 5 & 4 & 3 \\ 5 & 7 & 9 \end{pmatrix}_{2\times 3}$

c) $A^T \cdot B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 0 \end{pmatrix}_{3\times 2} \times \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}_{2\times 2} = \begin{pmatrix} 7 & 4 \\ 5 & 5 \\ 3 & 6 \end{pmatrix}_{3\times 2}$

d) $A + 2C = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \end{pmatrix}_{2\times 3} + \begin{pmatrix} 2 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix}_{2\times 3} = \begin{pmatrix} 3 & 2 & 5 \\ 4 & 3 & 0 \end{pmatrix}_{2\times 3}$

Solve simultaneous linear equation using matrices

① solve $2x + y = 4$ and $x + 3y = 7$:-

$A^{-1} \cdot \underbrace{\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}}_{A} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = A^{-1} \cdot \begin{pmatrix} 4 \\ 7 \end{pmatrix}$

$A^{-1} = \frac{1}{|A|} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{(2\times 3) - 1} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} = 0.2 \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.2 \\ -0.2 & 0.4 \end{pmatrix}_{2\times 2}$

$= \begin{pmatrix} 3/5 & -1/5 \\ -1/5 & 2/5 \end{pmatrix}_{2\times 2}$

$I \times \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 3/5 & -1/5 \\ -1/5 & 2/5 \end{pmatrix}_2 \cdot \begin{pmatrix} 4 \\ 7 \end{pmatrix}$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{2}{3}\times 1 + \left(-\frac{1}{3}\right)\times 7 \\ \left(-\frac{1}{3}\right)\times 1 + \left(\frac{2}{3}\right)\times 7 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

02. $2x - y = 1$ and $x + y = 5$ -:

$$* \underbrace{\begin{pmatrix} 2 & (-1) \\ 1 & 1 \end{pmatrix}_{2\times 2}}_{A} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

$$|A| = 2\times 1 - (-1)\times 1 = 3$$

$$\therefore A^{-} = \frac{1}{3}\begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

$$\therefore A^{-} \cdot A \begin{pmatrix} x \\ y \end{pmatrix} = A^{-} \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

$$I \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}\begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

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$2x + 3y = 13$
$3x + y = 9$

if above equations are represented by $A \cdot \begin{bmatrix} x \\ y \end{bmatrix}_{2\times 1} = B$

01. find A :- $A = \begin{pmatrix} 2 & 3 \\ 3 & 1 \end{pmatrix}_{2\times 2}$

02. find B :- $B = \begin{pmatrix} 13 \\ 9 \end{pmatrix}_{2\times 1}$

03. find $|A|$:- $|A| = 2\times 1 - 3\times 3 = (-7)$

04. find A^{-1} :- $A^{-1} = \frac{1}{|A|}\begin{pmatrix} 1 & -3 \\ -3 & 2 \end{pmatrix} = \frac{1}{-7}\begin{pmatrix} 1 & -3 \\ -3 & 2 \end{pmatrix}$

$= \begin{pmatrix} (-1/7) & 3/7 \\ 3/7 & (-2/7) \end{pmatrix}_{2\times 2}$

$1/-7$

$-1/7$

$-3/-7$

$= 3/7$

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01 Solve $2x + y = 8$, $x - 2y = (-1)$ using matrices

$$\begin{pmatrix} 2 & 1 \\ 1 & (-2) \end{pmatrix}_{2\times2} \cdot \begin{pmatrix} x \\ y \end{pmatrix}_{2\times1} = \begin{pmatrix} 8 \\ -1 \end{pmatrix}_{2\times1}$$

$$A \cdot \begin{pmatrix} x \\ y \end{pmatrix}_{2\times1} = B$$

$$A^{-1} \cdot A \cdot \begin{pmatrix} x \\ y \end{pmatrix} = A^{-1} \cdot B$$

$$I \cdot \begin{pmatrix} x \\ y \end{pmatrix}_{2\times1} = A^{-1} \cdot \begin{pmatrix} 8 \\ 1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{|A|}\begin{pmatrix} -2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{(2\times(-2) - 1\times1)}\begin{pmatrix} -2 & -1 \\ -1 & 2 \end{pmatrix}$$

$$= -\frac{1}{5}\begin{pmatrix} -2 & -1 \\ -1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2/5 & 1/5 \\ 1/5 & -2/5 \end{pmatrix}_{2\times2}$$

$$\begin{pmatrix} x \\ y \end{pmatrix}_{2\times2} = \begin{pmatrix} 2/5 & 1/5 \\ 1/5 & -2/5 \end{pmatrix}\begin{pmatrix} 8 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} 15/5 = 3 \\ 8/5 - 2 \end{pmatrix}_{2\times2}$$

$\therefore x = 3$, $y = 2$

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ex. $3x - y = 4$, $x + 2y = (-1)$, $A = ?$, $B = ?$, $|A| = ?$, $A^{-1} = ?$
$x = ?$, $y = ?$

$$\begin{pmatrix} 3 & (-1) \\ 1 & 2 \end{pmatrix}_{2\times 2} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}_{2\times 1}$$

$$A = \begin{pmatrix} 3 & (-1) \\ 1 & 2 \end{pmatrix}_{2\times 2} \quad , \quad B = \begin{pmatrix} 4 \\ -1 \end{pmatrix}_{2\times 1}$$

$$A \cdot \begin{pmatrix} x \\ y \end{pmatrix} = B \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$A^{-1} \cdot A \begin{pmatrix} x \\ y \end{pmatrix} = A^{-} \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$|A| = 3 \times 2 - (-1) \times 1 = 7$$

$$A^{-1} = \frac{1}{|A|} \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 2/7 & 1/7 \\ -1/7 & 3/7 \end{pmatrix}_{2\times 2}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2/7 & 1/7 \\ -1/7 & 3/7 \end{pmatrix} \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ -1 \end{pmatrix}_{2\times 1}$$

$\therefore x = 1$, $y = (-1)$